

Below is the **Normalization Foundation v1** for the PEK/MCP system.

The goal is:

Raw Dispensary Product Listing

- normalized DPL
- cannabis eligibility gate
- brand normalization
- product/form/size/mg extraction
- PEK generation
- hard rule matching
- MCP candidate creation
- review queue or auto-link
- price comparison index

The most important rule:

Minimize shopper categories.

Preserve exact product differences inside PEK fields.

So we do **not** create endless categories like Hash Hole, Blunt, AIO, Moonrock, Beverage, Capsule, etc. Those become **form, subform, infusion, extract, or hardware fields**.

1. Core object definitions

DPL — Dispensary Product Listing

A DPL is the raw product as shown by one dispensary.

It keeps store-specific information:

source dispensary

raw title

raw brand

raw category

price

sale price

menu URL

image URL
description
raw platform data

A DPL is **not** the canonical product. It is just a store-level sighting.

MCP — Master Canonical Product

An MCP is the clean product card representing the real-world sellable product.

Example:

MFNY 0.5g Hash Burger Live Resin Vape Cartridge

Multiple DPLs can connect to one MCP if they represent the same exact product.

PEK — Product Equality Key

The PEK is the product fingerprint used for matching.

It should describe the **exact sellable product**, not just a vague shopper category.

General PEK structure:

```
normalized_brand  
| normalized_category  
| normalized_form  
| subform  
| normalized_size  
| normalized_product_name  
| extract_type  
| infusion_type  
| hardware_type  
| count  
| cannabinoid_profile  
| ratio
```

Blank fields stay blank. We do **not** fabricate missing facts.

2. Pipeline overview

Step 1 — Ingest raw menus

Every product row from a dispensary menu enters a raw staging table exactly as received.

Do not clean destructively at this stage.

Schema: **raw_dispensary_product_listings**

```
CREATE TABLE raw_dispensary_product_listings (  
  raw_dpl_id TEXT PRIMARY KEY,  
  batch_id TEXT NOT NULL,  
  
  source_dispensary TEXT NOT NULL,  
  source_menu_name TEXT,  
  source_platform TEXT,  
  
  source_product_title TEXT NOT NULL,  
  source_brand TEXT,  
  source_category TEXT,  
  source_subcategory TEXT,  
  
  price NUMERIC,  
  sale_price NUMERIC,  
  currency TEXT DEFAULT 'USD',  
  
  thc_raw TEXT,  
  cbd_raw TEXT,  
  weight_raw TEXT,  
  size_raw TEXT,  
  count_raw TEXT,  
  
  image_url_if_available TEXT,  
  product_url_if_available TEXT,  
  description TEXT,  
  
  raw_payload JSONB,  
  first_seen TIMESTAMP,  
  last_seen TIMESTAMP,
```

```
    ingest_status TEXT DEFAULT 'ingested'  
);
```

Purpose:

Keep source truth untouched.
All later normalization should be reproducible.

3. Pre-normalization cleaning

Before extracting fields, clean the title and text fields.

Text normalization rules

Apply to title, brand, category, description:

- trim whitespace
- collapse repeated spaces
- normalize apostrophes
- normalize dashes
- remove repeated separators
- preserve meaningful separators like |, -, :, /
- standardize unicode characters
- standardize case for matching but preserve display case separately

Examples:

"Veteran's Choice - Blue Dream 1G Cart"
→ display title: "Veteran's Choice - Blue Dream 1G Cart"
→ search title: "veterans choice blue dream 1g cart"

Remove noisy menu decorations

Strip or ignore:

SALE

NEW
Staff Pick
Best Seller
Limited
BOGO
Online Only
Vendor Day
Fresh Drop

Do not remove these if they are part of the product name, but normally they are menu labels.

4. Cannabis eligibility gate

Before creating MCPs, determine whether the listing is eligible for cannabis price comparison.

Eligible cannabis categories

Use minimized top-level categories:

Flower
Pre-Rolls
Vapes
Concentrates
Edibles
Topicals
Tinctures

Optional later:

CBD / Hemp
Medical

But for now, keep the core cannabis categories tight.

Excluded from cannabis MCP matching

These can remain in raw inventory but should not create cannabis MCPs or PEKs:

- Accessories
- Merchandise
- Apparel
- Glassware
- Rolling Papers
- Wraps
- Cones
- Batteries
- Chargers
- Lighters
- Torches
- Grinders
- Trays
- Storage Containers
- Stash Jars
- Dab Tools
- Cleaning Supplies
- Devices not containing cannabis

Rule

If listing is accessory/merch/non-cannabis:

- keep raw DPL
- create normalized DPL with comparison_status = excluded_non_cannabis
- do not create cannabis PEK
- do not create cannabis MCP
- do not enter cannabis review queue
- do not learn cannabis brand/product aliases

Field: comparison_status

- eligible_cannabis
- excluded_accessory
- excluded_merchandise
- excluded_non_cannabis
- needs_review
- bad_source_data

5. Category normalization

Use broad shopper/menu buckets.

Canonical category map

Raw/menu concept	Normalized category
Flower, Bud, Small Buds, Smalls, Shake, Ground Flower	Flower
Pre-roll, Preroll, Joint, Blunt, Infused Pre-Roll, Hash Hole	Pre-Rolls
Vape, Cart, Cartridge, Disposable, AIO, Pod	Vapes
Concentrate, Rosin, Resin, Badder, Sauce, Diamonds, Hash, RSO	Concentrates
Edible, Gummies, Chocolate, Beverage, Capsule, Tablet, Lozenge, Oral	Edibles
Topical, Balm, Lotion, Salve	Topicals
Tincture, Drops	Tinctures
Battery, Paper, Glass, Lighter, Grinder, Tray	Excluded accessory
Shirt, Hat, Hoodie, Merch	Excluded merchandise

6. Form and subform normalization

Categories stay broad. Forms and subforms carry product detail.

Flower

Title/source keyword	Category	Normalized form	Subform
flower, bud	Flower	Flower	
eighth, 3.5g	Flower	Flower	Eighth
smalls, small buds	Flower	Flower	Small Buds

shake	Flower	Flower	Shake
ground flower	Flower	Flower	Ground Flower
moonrock, moon rock as loose flower	Flower	Flower	Moonrocks

Pre-Rolls

Title/source keyword	Category	Normalized form	Subform
pre-roll, preroll, joint	Pre-Rolls	Pre-Roll	
blunt	Pre-Rolls	Pre-Roll	Blunt
hash hole, hash-hole, hashhole	Pre-Rolls	Pre-Roll	Hash Hole
infused pre-roll	Pre-Rolls	Pre-Roll	Infused
dogwalker	Pre-Rolls	Pre-Roll	Dogwalker
mini pre-roll	Pre-Rolls	Pre-Roll	Mini
moonrock pre-roll	Pre-Rolls	Pre-Roll	Moonrock

Important:

Infused Pre-Roll is not a top-level category.

It is category = Pre-Rolls, normalized_form = Pre-Roll, infusion_type = Infused.

Vapes

Title/source keyword	Category	Normalized form	Hardware type
cart, cartridge, 510	Vapes	Vape Cartridge	510 Cartridge
disposable	Vapes	Disposable Vape	Disposable
AIO, all-in-one	Vapes	Disposable Vape	AIO
pod	Vapes	Vape Pod	Pod
starter kit	Vapes	Vape Kit	Kit

Important:

AIO, Cartridge, Pod, Disposable are vape keyphrases/hardware types.
They help define PEK identity but do not become top-level categories.

Concentrates

Title/source keyword	Category	Normalized form	Extract type
live resin	Concentrate s	Concentrate	Live Resin
cured resin	Concentrate s	Concentrate	Cured Resin
resin	Concentrate s	Concentrate	Resin
live rosin	Concentrate s	Concentrate	Live Rosin
rosin	Concentrate s	Concentrate	Rosin
badder, batter	Concentrate s	Concentrate	Badder
budder	Concentrate s	Concentrate	Budder
sauce	Concentrate s	Concentrate	Sauce
diamonds	Concentrate s	Concentrate	Diamonds
crumble	Concentrate s	Concentrate	Crumble
hash	Concentrate s	Concentrate	Hash
RSO, Rick Simpson Oil, RSO syringe	Concentrate s	Concentrate	RSO

Important:

RSO syringe always maps to Concentrates.
Syringe can be stored as subform/hardware/package style.

Edibles

Top-level category:

Edibles

Subforms:

Title/source keyword	Category	Normalized form
gummy, gummies	Edibles	Gummy
chocolate	Edibles	Chocolate
beverage, drink, seltzer, lemonade, tea	Edibles	Beverage
capsule	Edibles	Capsule
tablet	Edibles	Tablet
lozenge	Edibles	Lozenge
oral	Edibles	Oral
baked good, cookie, brownie	Edibles	Baked Good

Important:

Beverage and oral products live under Edibles.
They remain distinct through normalized_form/subform.

7. Brand normalization

Brand normalization must happen **before** PEK generation.

Brand normalization flow

raw_brand

- clean formatting
- check approved brand alias table
- check title prefix if raw brand is missing
- assign normalized_brand
- assign brand_confidence

Brand alias levels

Level 1 — automatic formatting normalization

Safe changes:

case normalization
apostrophe normalization
spacing cleanup
accent normalization
common punctuation cleanup

Examples:

OFF HOURS → Off Hours
STIIIZY → Stiizy
House Of Sacci → House of Sacci

Level 2 — approved aliases

Approved once, then reused.

Examples:

Veteran's Choice → Veterans Choice
Veterans Choice → Veterans Choice
Vet Choice → Veterans Choice
VCC → Veterans Choice
Veterans Choice Creations → Veterans Choice

ElectraLeaf → Electraleaf
Electraleaf NY → Electraleaf

Pearl by Gron → Pearls by Grön

Pearls by Gron → Pearls by Grön

6 Point Cannabis → 6 Points Cannabis

6 Points → 6 Points Cannabis

Level 3 — uncertain aliases

These go to review if they might be:

brand

product line

collab

store label

strain name

Schema: **brands**

```
CREATE TABLE brands (  
  brand_id TEXT PRIMARY KEY,  
  canonical_brand_name TEXT NOT NULL,  
  normalized_brand_key TEXT UNIQUE NOT NULL,  
  
  brand_status TEXT DEFAULT 'active',  
  notes TEXT,  
  
  created_at TIMESTAMP,  
  updated_at TIMESTAMP  
);
```

Schema: **brand_aliases**

```
CREATE TABLE brand_aliases (  
  brand_alias_id TEXT PRIMARY KEY,  
  
  raw_brand_value TEXT NOT NULL,  
  normalized_raw_value TEXT NOT NULL,  
  
  canonical_brand_id TEXT REFERENCES brands(branch_id),  
  canonical_brand_name TEXT NOT NULL,
```

```
alias_type TEXT,  
alias_confidence NUMERIC,  
  
approval_status TEXT NOT NULL,  
approved_by TEXT,  
approved_at TIMESTAMP,  
  
example_dpl_ids TEXT[],  
notes TEXT,  
  
created_at TIMESTAMP,  
updated_at TIMESTAMP  
);
```

Recommended `approval_status` values:

```
auto_normalized  
approved  
needs_review  
rejected  
deprecated
```

Recommended `alias_type` values:

```
case_variant  
punctuation_variant  
abbreviation  
misspelling  
legacy_name  
brand_line  
manual_alias
```

8. Brand extraction from title when brand is blank

If `source_brand` is blank and product is eligible cannabis, attempt title-based extraction.

Patterns

Brand | Product
Brand - Product
Brand: Product
Brand / Product

Examples:

New York Honey | Cured Resin SS13

→ Brand: New York Honey

→ Product: SS13

→ Extract type: Cured Resin

Ichi Roll | Uninfused 4pk | Afterglow Haze

→ Brand: Ichi Roll

→ Product: Afterglow Haze

→ Count: 4

→ Infusion type: None

Rule

Only extract title-prefix brand if:

- prefix matches known brand alias, or
- prefix appears repeatedly as a brand-like value, or
- source pattern strongly indicates brand separator.

If not confident:

normalized_brand = blank

comparison_status = needs_review

review_issue = missing_critical_data

9. Product name / strain normalization

The normalized product name should represent the strain, flavor, product line, or SKU name.

Remove from product name

Do not let these pollute the product name:

brand
size
weight
count
form
category
extract type
infusion type
hardware type
THC/CBD values
sale labels

Example:

Raw:

"MFNY Hash Burger Live Resin 510 Cartridge 0.5g"

Extract:

brand = MFNY
product_name = Hash Burger
extract_type = Live Resin
hardware_type = 510 Cartridge
normalized_form = Vape Cartridge
size = 0.5g

Preserve actual product identity words

Keep:

strain names
flavors
product lines
SKU names
collab names
ratio names when branded

Examples:

Blue Dream
Hash Burger

Watermelon Z
Pineapple Mango
Go
Chill
Sleep
Blue Raspberry
Apple Fritter

10. Product/strain/flavor alias system

This should grow over time.

Do **not** over-merge product names automatically.

Alias types

strain_alias
flavor_alias
product_line_alias
spelling_variant
punctuation_variant
abbreviation
brand_specific_alias

Examples

Likely safe after review:

Red Headed Stranger → Redheaded Stranger
Biscotti Pancake → Biscotti Pancakes
Lilac Diesel → Liliac Diesel

More dangerous:

Blue Razz → Blue Raspberry
Plant of the Grapes → Planet of the Grapes

These may need brand-specific approval.

Rule

Brand aliases can often be global.

Product/flavor aliases should usually be brand-scoped unless obviously identical.

Schema: **product_aliases**

```
CREATE TABLE product_aliases (  
  product_alias_id TEXT PRIMARY KEY,  
  
  alias_type TEXT NOT NULL,  
  
  raw_value TEXT NOT NULL,  
  normalized_raw_value TEXT NOT NULL,  
  canonical_value TEXT NOT NULL,  
  
  brand_scope_id TEXT REFERENCES brands(brand_id),  
  form_scope TEXT,  
  category_scope TEXT,  
  
  approval_status TEXT NOT NULL,  
  confidence NUMERIC,  
  
  example_dpl_ids TEXT[],  
  notes TEXT,  
  
  created_at TIMESTAMP,  
  updated_at TIMESTAMP  
);
```

Recommended **approval_status** values:

- approved_alias
- needs_review
- rejected_alias
- brand_specific_alias
- deprecated

11. Size normalization

Size means different things depending on category.

Flower / Pre-Rolls / Vapes / Concentrates

Normalize weight to grams.

Examples:

1/8 → 3.5g
eighth → 3.5g
3.5 grams → 3.5g
half gram → 0.5g
.5g → 0.5g
1 gram → 1g
2 gram → 2g
quarter → 7g
half ounce → 14g
ounce → 28g

Size table

Raw	Normalized
.5g	0.5g
0.5 gram	0.5g
half gram	0.5g
1 gram	1g
1g	1g
eighth	3.5g
1/8	3.5g
quarter	7g
1/4	7g
half oz	14g
ounce	28g

Pre-roll count and total weight

For pre-rolls, parse both:

individual unit size
count
total package weight

Examples:

5pk 0.5g
→ count = 5
→ unit_weight = 0.5g
→ total_weight = 2.5g
→ normalized_size = 2.5g
→ package_descriptor = 5pk x 0.5g

4pk 1g
→ count = 4
→ unit_weight = 1g
→ total_weight = 4g

If title only says:

5pk

Then:

count = 5
normalized_size = blank unless total weight is stated

Do not invent total grams.

12. Edible milligram normalization

For edibles, size is not flower-style weight.

Use:

total package cannabinoid mg
serving mg
count
ratio
cannabinoid profile

Edible parser should extract

package_thc_mg
serving_thc_mg
package_cbd_mg
serving_cbd_mg
package_cbn_mg
serving_cbn_mg
package_cbg_mg
serving_cbg_mg
count
ratio
cannabinoid_profile

Ignore Dutchie gram artifacts

For edibles and beverages, ignore these unless title confirms real package weight:

.01g
.1g
0.01g
0.1g

These are usually platform artifacts, not product size.

Edible mg patterns

Raw title pattern

Extraction

100mg

package_thc_mg = 100, if THC implied

100mg THC

package_thc_mg = 100

10pk 100mg	count = 10, package_thc_mg = 100
10ct 100mg	count = 10, package_thc_mg = 100
10 x 10mg	count = 10, serving_thc_mg = 10, package_thc_mg = 100
10 gummies 10mg each	count = 10, serving_thc_mg = 10, package_thc_mg = 100
5mg Beverage	package_thc_mg = 5 if single-serve
2.5mg THC : 2.5mg CBD	serving_thc_mg = 2.5, serving_cbd_mg = 2.5
1:1 THC:CBD	ratio = 1:1, cannabinoid_profile = THC/CBD
THC:CBN	cannabinoid_profile = THC/CBN

New York adult-use edible rule

Use this as validation, not invention:

Expected adult-use edible max:

serving THC <= 10mg

package THC <= 100mg

Rule:

Do not auto-fill 100mg just because a NY edible is missing mg.

Use 100mg as a ceiling / validation rule only.

Edible normalized size examples

100mg THC

10mg THC

100mg THC / 100mg CBD

20pk / 100mg THC

10pk x 10mg THC

5mg THC

13. Extract type normalization

Extract type is matching-critical.

Extract type values

Distillate
Live Resin
Cured Resin
Resin
Live Rosin
Rosin
Liquid Diamonds
Diamonds
Sauce
Badder
Budder
Crumble
Hash
Bubble Hash
Kief
RSO
Unknown

Alias examples

batter → Badder
badder → Badder
live resin sauce → Live Resin + Sauce
live rosin → Live Rosin
LR → Live Resin only if brand/title context supports it

Hard rule

Unknown extract_type does not equal known extract_type.

So:

Blue Dream 1g Cartridge

does not automatically match:

Blue Dream 1g Live Resin Cartridge

14. Infusion type normalization

Infusion type is especially important for pre-rolls.

Infusion values

None
Infused
Hash Infused
Kief Infused
Live Resin Infused
Live Rosin Infused
Rosin Infused
Diamond Infused
Moonrock Infused
Terpene Infused
Unknown

Rules

Uninfused $\neq$ Infused
Hash Infused $\neq$ Kief Infused unless approved
Live Resin Infused $\neq$ Live Rosin Infused
Generic Infused $\neq$ specific infusion type unless brand/product confirms equivalence

Hash hole rule

hash hole / hash-hole / hashhole
→ category = Pre-Rolls
→ normalized_form = Pre-Roll
→ subform = Hash Hole
→ infusion_type = Hash Infused, Rosin Infused, Live Rosin Infused, or Infused depending title

15. Hardware type normalization

Mainly for vapes.

Hardware values

510 Cartridge

Disposable

AIO

Pod

Vape Kit

Unknown

Rules

AIO = All-In-One Disposable Vape

Cart / Cartridge / 510 = Vape Cartridge

Pod = Vape Pod

Do not merge different hardware types.

These are separate MCPs:

Jaunty Blue Dream 1g Vape Cartridge

Jaunty Blue Dream 1g AIO Disposable Vape

Jaunty Blue Dream 1g Vape Pod

16. Dominance / type normalization

Useful but not always matching-critical.

Values

Indica

Sativa

Hybrid

Indica Hybrid

Sativa Hybrid

CBD
THC
Balanced
Unknown

Rule

Use dominance/type as supporting evidence.

Do not block a match only because one menu says Hybrid and another omits it.

But flag contradictions:

Same PEK candidate:
one says Indica
one says Sativa
→ dominance_conflict review

17. Normalized DPL schema

This is the cleaned row-level object before MCP grouping.

```
CREATE TABLE normalized_dispensary_product_listings (  
  dpl_id TEXT PRIMARY KEY,  
  raw_dpl_id TEXT REFERENCES raw_dispensary_product_listings(raw_dpl_id),  
  batch_id TEXT NOT NULL,  
  
  source_dispensary TEXT NOT NULL,  
  source_menu_name TEXT,  
  source_product_title TEXT NOT NULL,  
  normalized_title TEXT,  
  
  source_brand TEXT,  
  normalized_brand TEXT,  
  brand_id TEXT,  
  brand_confidence NUMERIC,  
  
  raw_category TEXT,  
  normalized_category TEXT,  
  normalized_form TEXT,
```

subform TEXT,

product_name_or_strain TEXT,
normalized_product_name TEXT,

size_value NUMERIC,
size_unit TEXT,
normalized_size TEXT,

unit_weight_value NUMERIC,
unit_weight_unit TEXT,
total_weight_value NUMERIC,
total_weight_unit TEXT,

count INTEGER,

package_thc_mg NUMERIC,
serving_thc_mg NUMERIC,
package_cbd_mg NUMERIC,
serving_cbd_mg NUMERIC,
package_cbn_mg NUMERIC,
serving_cbn_mg NUMERIC,
package_cbg_mg NUMERIC,
serving_cbg_mg NUMERIC,

cannabinoid_profile TEXT,
ratio TEXT,

extract_type TEXT,
infusion_type TEXT,
hardware_type TEXT,
dominance_or_type TEXT,

thc_value TEXT,
cbd_value TEXT,

price NUMERIC,
sale_price NUMERIC,
effective_price NUMERIC,

image_url_if_available TEXT,
product_url_if_available TEXT,
description TEXT,

```
comparison_status TEXT,  
proposed_pek TEXT,  
  
extraction_confidence NUMERIC,  
source_notes TEXT,  
  
created_at TIMESTAMP,  
updated_at TIMESTAMP  
);
```

18. PEK generation

Only generate PEK for cannabis-eligible DPLs.

General PEK

```
normalized_brand  
| normalized_category  
| normalized_form  
| subform  
| normalized_size  
| normalized_product_name  
| extract_type  
| infusion_type  
| hardware_type  
| count  
| cannabinoid_profile  
| ratio
```

Flower PEK

Brand | Flower | Form/Subform | Size | Strain/Product Name

Example:

Veterans Choice | Flower | Small Buds | 7g | Apple Fritter

Pre-Roll PEK

Brand | Pre-Rolls | Pre-Roll | Subform | Total Size | Count | Strain/Product Name | Infusion Type

Example:

Ichu Roll | Pre-Rolls | Pre-Roll | | | 4 | Afterglow Haze | None

Hash hole example:

Brand | Pre-Rolls | Pre-Roll | Hash Hole | 2g | 1 | Strain | Live Rosin Infused

Vape PEK

Brand | Vapes | Normalized Form | Hardware Type | Size | Product/Strain | Extract Type

Example:

MFNY | Vapes | Vape Cartridge | 510 Cartridge | 0.5g | Hash Burger | Live Resin

Concentrate PEK

Brand | Concentrates | Concentrate | Size | Product/Strain | Extract Type | Subform

Example:

New York Honey | Concentrates | Concentrate | 1g | SS13 | Cured Resin |

Edible PEK

Brand | Edibles | Normalized Form | Flavor/Product Line | Package MG | Count | Cannabinoid Profile | Ratio

Example:

Off Hours | Edibles | Gummy | Blue Raspberry | 100mg THC | 10 | THC |

Beverage example:

Ayrloom | Edibles | Beverage | Honeycrisp | 10mg THC | 1 | THC/CBD | 1:1

19. Hard rule gates before matching

False positives are worse than missed matches.

Before fuzzy matching or embeddings, apply hard gates.

Gate 1 — cannabis eligibility

Excluded accessory/merch/non-cannabis cannot match cannabis MCP.

Gate 2 — brand compatibility

Brands must match exactly after alias normalization.

Allowed:

Veteran's Choice = Veterans Choice

Vet Choice = Veterans Choice

Not allowed:

Similar-looking but unapproved brand names

Gate 3 — category compatibility

Categories must match after minimized category normalization.

Allowed:

Beverage → Edibles

Capsule → Edibles

Blunt → Pre-Rolls

Hash Hole → Pre-Rolls

RSO → Concentrates

Not allowed:

Flower ≠ Concentrates

Vapes ≠ Concentrates

Pre-Rolls ≠ Flower

Gate 4 — form compatibility

Forms must match where form is identity-critical.

Examples:

Vape Cartridge ≠ Disposable Vape

Vape Pod ≠ Vape Cartridge

Gummy ≠ Beverage

Capsule ≠ Chocolate

Even though beverages and gummies both live under Edibles, they are not the same form.

Gate 5 — size compatibility

Usually size must match.

Examples:

0.5g ≠ 1g

3.5g ≠ 7g

10mg ≠ 100mg

5pk ≠ 10pk

If size is missing on one side, send to review rather than auto-match.

Gate 6 — product/strain/flavor compatibility

Must match or be an approved alias.

Do not merge different strains/flavors.

Examples:

Blue Dream ≠ Sour Diesel

Blue Raspberry ≠ Watermelon

Hash Burger ≠ Honey Banana

Gate 7 — extract compatibility

Must match for:

vapes

concentrates

infused pre-rolls

Hard no:

Live Resin ≠ Live Rosin

Live Resin ≠ Distillate

Rosin ≠ Resin

Badder ≠ Sauce

Diamonds ≠ Hash

Gate 8 — infusion compatibility

Must match for pre-rolls and infused products.

Hard no:

Uninfused ≠ Infused

Hash Infused ≠ Kief Infused

Live Resin Infused ≠ Live Rosin Infused

Gate 9 — hardware compatibility

For vapes:

Cartridge ≠ Disposable ≠ AIO ≠ Pod

Gate 10 — ratio/cannabinoid compatibility

For edibles:

THC ≠ THC/CBD

1:1 ≠ 2:1

THC/CBN ≠ THC-only

20. Matching confidence framework

After hard gates pass, score candidate matches.

Suggested scoring

Signal	Weight
Brand exact/approved alias	Required
Category match	Required
Form match	Required
Size/mg/count match	High
Product/strain/flavor match	High
Extract/infusion match	High where relevant
Hardware match	High for vapes
Ratio/cannabinoid profile match	High for edibles
Title similarity	Medium
Description similarity	Low/medium
Image similarity	Supporting only
THC/CBD lab value similarity	Low/supporting

Confidence bands

95–100 = auto-match

80–94 = match but needs review
60–79 = candidate only
below 60 = provisional new MCP

But even a high score cannot override a hard gate conflict.

21. MCP creation

When no safe existing MCP match exists, create a provisional MCP.

Canonical title structure

Use:

Brand + Size/MG/Count + Product Name + Modifier + Form

But keep titles readable.

Examples:

MFNY 0.5g Hash Burger Live Resin Vape Cartridge
Off Hours 100mg Blue Raspberry Gummies 10pk
Ayrloom 10mg Honeycrisp 1:1 THC:CBD Beverage
Veterans Choice 7g Apple Fritter Small Buds Flower
Ichi Roll 4pk Afterglow Haze Pre-Rolls
New York Honey 1g SS13 Cured Resin Concentrate

MCP schema

```
CREATE TABLE master_canonical_products (  
  mcp_id TEXT PRIMARY KEY,  
  
  pek TEXT UNIQUE,  
  canonical_title TEXT NOT NULL,  
  search_title TEXT,  
  
  brand_id TEXT REFERENCES brands(brand_id),  
  normalized_brand TEXT NOT NULL,
```

normalized_category TEXT NOT NULL,
normalized_form TEXT NOT NULL,
subform TEXT,

canonical_product_name TEXT NOT NULL,

size_value NUMERIC,
size_unit TEXT,
normalized_size TEXT,

unit_weight_value NUMERIC,
total_weight_value NUMERIC,
count INTEGER,

package_thc_mg NUMERIC,
serving_thc_mg NUMERIC,
package_cbd_mg NUMERIC,
serving_cbd_mg NUMERIC,
package_cbn_mg NUMERIC,
serving_cbn_mg NUMERIC,
package_cbg_mg NUMERIC,
serving_cbg_mg NUMERIC,

cannabinoid_profile TEXT,
ratio TEXT,

extract_type TEXT,
infusion_type TEXT,
hardware_type TEXT,
dominance_or_type TEXT,

canonical_image_id TEXT,
canonical_image_url TEXT,
image_status TEXT,

confidence_score NUMERIC,
review_status TEXT,

first_seen TIMESTAMP,
last_seen TIMESTAMP,

created_at TIMESTAMP,
updated_at TIMESTAMP

);

Recommended **review_status** values:

approved
provisional
needs_review
rejected_duplicate
deprecated

22. Linking DPLs to MCPs

Once a DPL is matched to an MCP, store the relationship.

Schema: **mcp_dpl_links**

```
CREATE TABLE mcp_dpl_links (  
  link_id TEXT PRIMARY KEY,  
  
  mcp_id TEXT REFERENCES master_canonical_products(mcp_id),  
  dpl_id TEXT REFERENCES normalized_dispensary_product_listings(dpl_id),  
  
  match_confidence NUMERIC,  
  match_method TEXT,  
  match_reasons TEXT[],  
  
  needs_review BOOLEAN DEFAULT FALSE,  
  review_status TEXT,  
  
  linked_at TIMESTAMP,  
  updated_at TIMESTAMP  
);
```

Recommended **match_method** values:

exact_pek
approved_alias
high_confidence_match
manual_review

provisional_new_mcp

23. Review queue

Every uncertain case should be reviewable.

Review issue types

possible_duplicate
possible_same_product
brand_alias_uncertain
strain_conflict
flavor_conflict
size_conflict
form_conflict
category_conflict
dominance_conflict
extract_type_conflict
infusion_type_conflict
hardware_conflict
ratio_conflict
missing_critical_data
bad_source_data
price_outlier
excluded_non_cannabis

Schema: **product_review_queue**

```
CREATE TABLE product_review_queue (  
  review_id TEXT PRIMARY KEY,  
  batch_id TEXT,  
  
  issue_type TEXT NOT NULL,  
  risk_level TEXT NOT NULL,  
  
  dpl_ids TEXT[],  
  candidate_mcp_ids TEXT[],  
  
  source_titles_compared TEXT[],
```

```
reason_for_review TEXT,  
recommended_action TEXT,  
  
status TEXT DEFAULT 'open',  
reviewed_by TEXT,  
reviewed_at TIMESTAMP,  
resolution_notes TEXT,  
  
created_at TIMESTAMP,  
updated_at TIMESTAMP  
);
```

Recommended `risk_level` values:

low
medium
high
critical

24. Rejected near-matches

This is important because it teaches the system what **not** to merge.

Examples

Same brand + strain + size, but cartridge vs disposable
Same brand + strain + size, but live resin vs distillate
Same brand + flavor + mg, but 1:1 vs THC-only
Same brand + strain, but 3.5g flower vs 7g small buds

Schema: `rejected_near_matches`

```
CREATE TABLE rejected_near_matches (  
  rejected_match_id TEXT PRIMARY KEY,  
  
  dpl_id_1 TEXT,  
  dpl_id_2 TEXT,  
  mcp_id_1 TEXT,
```

```
mcp_id_2 TEXT,  
  
title_1 TEXT,  
title_2 TEXT,  
  
why_they_look_similar TEXT,  
why_they_should_not_match TEXT,  
  
rule_learned TEXT,  
confidence NUMERIC,  
  
created_at TIMESTAMP  
);
```

25. Data quality flags

These are not always matching problems, but they should be tracked.

Common flags

```
missing_brand  
missing_size  
missing_mg  
missing_count  
bad_edible_weight_artifact  
conflicting_category  
conflicting_form  
conflicting_extract_type  
conflicting_infusion_type  
possible_wrong_source_category  
possible_wrong_brand  
price_outlier  
duplicate_listing  
title_too_generic
```

Schema: **data_quality_flags**

```
CREATE TABLE data_quality_flags (  
  flag_id TEXT PRIMARY KEY,
```

```
batch_id TEXT,  
  
dpl_id TEXT,  
source_dispensary TEXT,  
  
problem_type TEXT NOT NULL,  
problem_description TEXT,  
suggested_fix TEXT,  
  
confidence NUMERIC,  
status TEXT DEFAULT 'open',  
  
created_at TIMESTAMP,  
updated_at TIMESTAMP  
);
```

26. Price comparison index

Only eligible cannabis MCPs should enter price comparison.

Rule

A DPL enters price comparison only if:
comparison_status = eligible_cannabis
linked to valid MCP
product has usable price or sale price
product is in stock / active if stock data exists

Schema: **price_comparison_index**

```
CREATE TABLE price_comparison_index (  
  price_index_id TEXT PRIMARY KEY,  
  
  mcp_id TEXT REFERENCES master_canonical_products(mcp_id),  
  dpl_id TEXT REFERENCES normalized_dispensary_product_listings(dpl_id),  
  
  source_dispensary TEXT,  
  source_menu_name TEXT,
```

```
canonical_title TEXT,  
normalized_brand TEXT,  
normalized_category TEXT,  
normalized_form TEXT,  
normalized_size TEXT,  
  
price NUMERIC,  
sale_price NUMERIC,  
effective_price NUMERIC,  
  
product_url TEXT,  
image_url_from_store TEXT,  
  
in_stock BOOLEAN,  
last_seen TIMESTAMP,  
last_checked TIMESTAMP,  
  
created_at TIMESTAMP,  
updated_at TIMESTAMP  
);
```

27. Recommended processing order

This is the full operational flow.

1. Ingest raw menu rows.
2. Store untouched source data in raw DPL table.
3. Clean title/brand/category text.
4. Detect accessory/merch/non-cannabis exclusions.
5. Normalize broad category.
6. Normalize form, subform, hardware, extract, infusion.
7. Normalize brand using brand alias table.
8. If brand missing, attempt title-prefix extraction.
9. Extract product name / strain / flavor.
10. Normalize product aliases if approved.
11. Extract size:
 - flower/pre-roll/vape/concentrate = grams/count
 - edibles = mg/count/ratio/cannabinoid profile
12. Ignore edible gram artifacts.
13. Generate normalized DPL row.
14. If excluded_non_cannabis, stop cannabis matching.

15. Generate PEK for eligible cannabis row.
 16. Search existing MCPs by PEK and alias candidates.
 17. Apply hard gates.
 18. Score candidate matches.
 19. Auto-link, review, or create provisional MCP.
 20. Create rejected near-match records where useful.
 21. Update brand/product alias suggestions.
 22. Update price comparison index.
 23. Export review queue and data quality flags.
-

28. Current resolved system rules

These are now foundation rules:

Accessories and merchandise are excluded from cannabis MCP price comparison.

Hash hole maps to Pre-Rolls category.

Hash hole is an infused pre-roll subform/identity marker.

Blunt maps to Pre-Rolls category.

Blunt is a subform, not a category.

Moonrocks map according to sellable form:

moonrock pre-roll/blunt/joint → Pre-Rolls

loose moonrock flower → Flower

RSO syringe maps to Concentrates.

AIO maps to Vapes.

AIO is hardware/subform for Disposable Vape.

Cartridge, pod, disposable, AIO are vape identity fields.

Beverage, oral, capsule, tablet, lozenge all live under Edibles.

Edibles normalize by milligrams/count/ratio, not gram weight.

NY edible 100mg package max is a validation rule, not an automatic inferred size.

Brand aliases are required and must normalize before PEK generation.

Extract and infusion conflicts block automatic matching.

Different strains/flavors should not merge unless approved alias exists.

Different sizes/counts should not merge.

Unknown extract type does not equal known extract type.

29. Minimal category philosophy

Use this:

Flower
Pre-Rolls
Vapes
Concentrates
Edibles
Topicals
Tinctures
Accessories - excluded
Merchandise - excluded

Avoid this:

Hash Hole as category
Blunt as category
AIO as category
Beverage as top-level category
Capsule as top-level category
Moonrock as top-level category
RSO as top-level category

The product identity still remains precise because PEK uses:

normalized_form
subform
extract_type
infusion_type
hardware_type

size
count
mg
ratio
product name

That is the foundation: **fewer categories, stronger PEK fields.**